

Class 14

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Background

We are going to dive deeper into some genes and proteins to gain novel insights. For our lab today we are focusing on the HOXA1 gene

Data Import

Our data comes from:

Trapnell C, Hendrickson DG, Sauvageau M, Goff L et al. “*Differential analysis of gene regulation at transcript resolution with RNA-seq*”. Nat Biotechnol 2013 Jan;31(1):46-53. [PMID: 23222703](#)

```
library(DESeq2)
```

```
Loading required package: S4Vectors
```

```
Loading required package: stats4
```

Loading required package: BiocGenerics

Loading required package: generics

Attaching package: 'generics'

The following objects are masked from 'package:base':

as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
setequal, union

Attaching package: 'BiocGenerics'

The following objects are masked from 'package:stats':

IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

anyDuplicated, aperm, append, as.data.frame, basename, cbind,
colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
unsplit, which.max, which.min

Attaching package: 'S4Vectors'

The following object is masked from 'package:utils':

findMatches

The following objects are masked from 'package:base':

expand.grid, I, unname

Loading required package: IRanges

Attaching package: 'IRanges'

The following object is masked from 'package:grDevices':

windows

Loading required package: GenomicRanges

Loading required package: Seqinfo

Loading required package: SummarizedExperiment

Loading required package: MatrixGenerics

Loading required package: matrixStats

Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
colWeightedMeans, colWeightedMedians, colWeightedSds,
colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
rowWeightedSds, rowWeightedVars

Loading required package: Biobase

Welcome to Bioconductor

Vignettes contain introductory material; view with
'browseVignettes()'. To cite Bioconductor, see
'citation("Biobase")', and for packages 'citation("pkgname")'.

Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

rowMedians

The following objects are masked from 'package:matrixStats':

anyMissing, rowMedians

```
metaFile<-"GSE37704_metadata.csv"
countFile<-"GSE37704_featurecounts.csv"
#Now our data is in lets look at it
colData<-read.csv(metaFile, row.names = 1)
head(colData)
```

```
              condition
SRR493366 control_sirna
SRR493367 control_sirna
SRR493368 control_sirna
SRR493369      hoxa1_kd
SRR493370      hoxa1_kd
SRR493371      hoxa1_kd
```

```
countData<- read.csv(countFile, row.names = 1)
head(countData)
```

	length	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370
ENSG00000186092	918	0	0	0	0	0
ENSG00000279928	718	0	0	0	0	0
ENSG00000279457	1982	23	28	29	29	28
ENSG00000278566	939	0	0	0	0	0
ENSG00000273547	939	0	0	0	0	0

ENSG00000187634	3214	124	123	205	207	212
	SRR493371					
ENSG00000186092	0					
ENSG00000279928	0					
ENSG00000279457	46					
ENSG00000278566	0					
ENSG00000273547	0					
ENSG00000187634	258					

#Q. Complete the code below to remove the troublesome first column from countData

```
#Removing first column
countData <- as.matrix(countData[,2:7])
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000186092	0	0	0	0	0	0
ENSG00000279928	0	0	0	0	0	0
ENSG00000279457	23	28	29	29	28	46
ENSG00000278566	0	0	0	0	0	0
ENSG00000273547	0	0	0	0	0	0
ENSG00000187634	124	123	205	207	212	258

#Q. Complete the code below to filter countData to exclude genes (i.e. rows) where we have 0

```
#Remove all 0 reads
countData = countData[rowSums(countData)>0, ]
head(countData)
```

	SRR493366	SRR493367	SRR493368	SRR493369	SRR493370	SRR493371
ENSG00000279457	23	28	29	29	28	46
ENSG00000187634	124	123	205	207	212	258
ENSG00000188976	1637	1831	2383	1226	1326	1504
ENSG00000187961	120	153	180	236	255	357
ENSG00000187583	24	48	65	44	48	64
ENSG00000187642	4	9	16	14	16	16

```
#Or
to.keep<-rowSums(countData) !=0
countData<- countData[to.keep,]
```

Setup DESeq object

```
dds= DESeqDataSetFromMatrix(countData = countData,
  colData = colData,
  design=~condition)
```

Warning in DESeqDataSet(se, design = design, ignoreRank): some variables in design formula are characters, converting to factors

```
dds<-DESeq(dds)
```

estimating size factors

estimating dispersions

gene-wise dispersion estimates

mean-dispersion relationship

final dispersion estimates

fitting model and testing

```
res<- results(dds)
summary(res)
```

```
out of 15975 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)      : 4349, 27%
LFC < 0 (down)    : 4396, 28%
outliers [1]      : 0, 0%
```

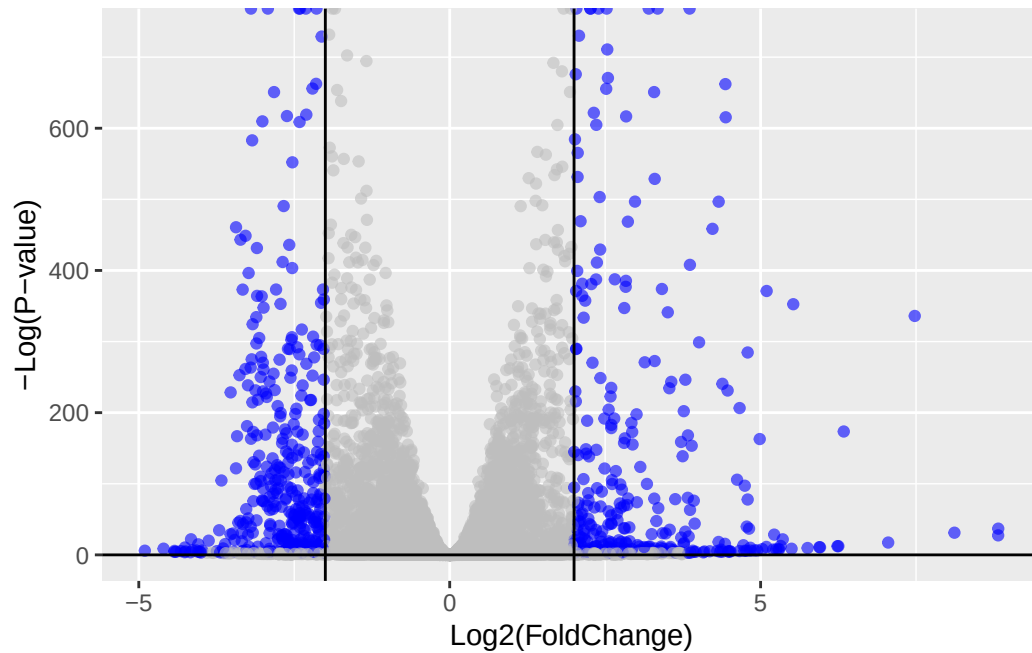
```
low counts [2]      : 1237, 7.7%  
(mean count < 0)  
[1] see 'cooksCutoff' argument of ?results  
[2] see 'independentFiltering' argument of ?results
```

```
#Q. Call the summary() function on your results to get a sense of how many genes are up or d  
#According to the summary function, 4349 are up regulated and 4396 are down regulated
```

Volcano Plot

```
library(ggplot2)  
#Q. Improve this plot by completing the below code, which adds color, axis labels and cutoff  
  
my_col<- rep("grey", nrow(res))  
my_col[abs(res$log2FoldChange)>2]<-"blue"  
my_col[res$padj>0.05]<-"grey"  
  
ggplot(res)+  
  aes(log2FoldChange, -log(padj))+  
  geom_point(alpha=0.6, col=my_col)+  
  geom_vline(xintercept = c(-2,2))+  
  geom_hline(yintercept=0)+  
  xlab("Log2(FoldChange)") +  
  ylab("-Log(P-value)")
```

Warning: Removed 1237 rows containing missing values or values outside the scale range (``geom_point()``).



Adding Gene Annotation

```
library(AnnotationDbi)
library(org.Hs.eg.db)
```

#Q. Use the mapIDs() function multiple times to add SYMBOL, ENTREZID and GENENAME annotation

```
#Lets look at the columns
columns(org.Hs.eg.db)
```

[1] "ACCNUM"	"ALIAS"	"ENSEMBL"	"ENSEMBLPROT"	"ENSEMBLTRANS"
[6] "ENTREZID"	"ENZYME"	"EVIDENCE"	"EVIDENCEALL"	"GENENAME"
[11] "GENETYPE"	"GO"	"GOALL"	"IPI"	"MAP"
[16] "OMIM"	"ONTOLOGY"	"ONTOLOGYALL"	"PATH"	"PFAM"
[21] "PMID"	"PROSITE"	"REFSEQ"	"SYMBOL"	"UCSCKG"
[26] "UNIPROT"				


```
res$symbol=mapIds(org.Hs.eg.db,
                  keys = row.names(res),
                  keytype = "ENSEMBL",
                  column = "SYMBOL",
                  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$entrez=mapIds(org.Hs.eg.db,
                  keys = row.names(res),
                  keytype = "ENSEMBL",
                  column = "ENTREZID",
                  multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
res$name=mapIds(org.Hs.eg.db,
                keys = row.names(res),
                keytype = "ENSEMBL",
                column = "GENENAME",
                multiVals = "first")
```

'select()' returned 1:many mapping between keys and columns

```
head(res,10)
```

log2 fold change (MLE): condition hoxa1 kd vs control sirna

Wald test p-value: condition hoxa1 kd vs control sirna

DataFrame with 10 rows and 9 columns

	baseMean	log2FoldChange	lfcSE	stat	pvalue
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>
ENSG00000279457	29.913579	0.1792571	0.3248216	0.551863	5.81042e-01
ENSG00000187634	183.229650	0.4264571	0.1402658	3.040350	2.36304e-03
ENSG00000188976	1651.188076	-0.6927205	0.0548465	-12.630158	1.43990e-36
ENSG00000187961	209.637938	0.7297556	0.1318599	5.534326	3.12428e-08
ENSG00000187583	47.255123	0.0405765	0.2718928	0.149237	8.81366e-01
ENSG00000187642	11.979750	0.5428105	0.5215598	1.040744	2.97994e-01
ENSG00000188290	108.922128	2.0570638	0.1969053	10.446970	1.51282e-25
ENSG00000187608	350.716868	0.2573837	0.1027266	2.505522	1.22271e-02

	padj	symbol	entrez	name
	<numeric>	<character>	<character>	<character>
ENSG00000188157	9128.439422			
ENSG00000237330	0.158192			
ENSG00000279457	6.86555e-01	NA	NA	NA
ENSG00000187634	5.15718e-03	SAMD11	148398	sterile alpha motif ..
ENSG00000188976	1.76549e-35	NOC2L	26155	NOC2 like nucleolar ..
ENSG00000187961	1.13413e-07	KLHL17	339451	kelch like family me..
ENSG00000187583	9.19031e-01	PLEKHN1	84069	pleckstrin homology ..
ENSG00000187642	4.03379e-01	PERM1	84808	PPARGC1 and ESRR ind..
ENSG00000188290	1.30538e-24	HES4	57801	hes family bHLH tran..
ENSG00000187608	2.37452e-02	ISG15	9636	ISG15 ubiquitin like..
ENSG00000188157	4.21963e-16	AGRN	375790	agrin
ENSG00000237330	NA	RNF223	401934	ring finger protein ..

#Q. Finally for this section let's reorder these results by adjusted p-value and save them to

```
#lets reorder
res= res[order(res$pvalue),]
write.csv(res,file = "deseq_results.csv")
```

Pathway Analysis

```
library(pathview)
```

```
#####
Pathview is an open source software package distributed under GNU General
Public License version 3 (GPLv3). Details of GPLv3 is available at
http://www.gnu.org/licenses/gpl-3.0.html. Particullary, users are required to
formally cite the original Pathview paper (not just mention it) in publications
or products. For details, do citation("pathview") within R.
```

The pathview downloads and uses KEGG data. Non-academic uses may require a KEGG license agreement (details at <http://www.kegg.jp/kegg/legal.html>).

```
#####
```

```
library(gage)
```

```
library(gageData)
data(kegg.sets.hs)
data(sigmet.idx.hs)
#Focusing only signaling and metabolic pathways
kegg.sets.hs=kegg.sets.hs[sigmet.idx.hs]
head(kegg.sets.hs,3)
```

```
$`hsa00232 Caffeine metabolism`
```

```
[1] "10" "1544" "1548" "1549" "1553" "7498" "9"
```

```
$`hsa00983 Drug metabolism - other enzymes`
```

```
[1] "10" "1066" "10720" "10941" "151531" "1548" "1549" "1551"
[9] "1553" "1576" "1577" "1806" "1807" "1890" "221223" "2990"
[17] "3251" "3614" "3615" "3704" "51733" "54490" "54575" "54576"
[25] "54577" "54578" "54579" "54600" "54657" "54658" "54659" "54963"
[33] "574537" "64816" "7083" "7084" "7172" "7363" "7364" "7365"
[41] "7366" "7367" "7371" "7372" "7378" "7498" "79799" "83549"
[49] "8824" "8833" "9" "978"
```

```
$`hsa00230 Purine metabolism`
```

```
[1] "100" "10201" "10606" "10621" "10622" "10623" "107" "10714"
[9] "108" "10846" "109" "111" "11128" "11164" "112" "113"
[17] "114" "115" "122481" "122622" "124583" "132" "158" "159"
[25] "1633" "171568" "1716" "196883" "203" "204" "205" "221823"
[33] "2272" "22978" "23649" "246721" "25885" "2618" "26289" "270"
[41] "271" "27115" "272" "2766" "2977" "2982" "2983" "2984"
[49] "2986" "2987" "29922" "3000" "30833" "30834" "318" "3251"
[57] "353" "3614" "3615" "3704" "377841" "471" "4830" "4831"
[65] "4832" "4833" "4860" "4881" "4882" "4907" "50484" "50940"
[73] "51082" "51251" "51292" "5136" "5137" "5138" "5139" "5140"
[81] "5141" "5142" "5143" "5144" "5145" "5146" "5147" "5148"
[89] "5149" "5150" "5151" "5152" "5153" "5158" "5167" "5169"
[97] "51728" "5198" "5236" "5313" "5315" "53343" "54107" "5422"
[105] "5424" "5425" "5426" "5427" "5430" "5431" "5432" "5433"
[113] "5434" "5435" "5436" "5437" "5438" "5439" "5440" "5441"
[121] "5471" "548644" "55276" "5557" "5558" "55703" "55811" "55821"
[129] "5631" "5634" "56655" "56953" "56985" "57804" "58497" "6240"
[137] "6241" "64425" "646625" "654364" "661" "7498" "8382" "84172"
[145] "84265" "84284" "84618" "8622" "8654" "87178" "8833" "9060"
[153] "9061" "93034" "953" "9533" "954" "955" "956" "957"
[161] "9583" "9615"
```

```
foldchanges<- res$log2FoldChange
names(foldchanges)=res$entrez
head(foldchanges)
```

```
      1266      54855      1465      2034      2150      6659
-2.422719  3.201955 -2.313738 -1.888019  3.344508  2.392288
```

```
#Lets see the results
keggres <- gage(foldchanges, gsets = kegg.sets.hs)
attributes(keggres)
```

```
$names
[1] "greater" "less"    "stats"
```

```
head(keggres$less)
```

	p.geomean	stat.mean	p.val
hsa04110 Cell cycle	8.995727e-06	-4.378644	8.995727e-06
hsa03030 DNA replication	9.424076e-05	-3.951803	9.424076e-05
hsa03013 RNA transport	1.375901e-03	-3.028500	1.375901e-03
hsa03440 Homologous recombination	3.066756e-03	-2.852899	3.066756e-03
hsa04114 Oocyte meiosis	3.784520e-03	-2.698128	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	8.961413e-03	-2.405398	8.961413e-03

	q.val	set.size	exp1
hsa04110 Cell cycle	0.001448312	121	8.995727e-06
hsa03030 DNA replication	0.007586381	36	9.424076e-05
hsa03013 RNA transport	0.073840037	144	1.375901e-03
hsa03440 Homologous recombination	0.121861535	28	3.066756e-03
hsa04114 Oocyte meiosis	0.121861535	102	3.784520e-03
hsa00010 Glycolysis / Gluconeogenesis	0.212222694	53	8.961413e-03

```
pathview(gene.data = foldchanges, pathway.id = "hsa04110")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/micha/OneDrive/Desktop/BIMM 143/Class 14

Info: Writing image file hsa04110.pathview.png

```
pathview(gene.data=foldchanges, pathway.id="hsa04110", kegg.native=FALSE)
```

'select()' returned 1:1 mapping between keys and columns

Warning: reconcile groups sharing member nodes!

```
      [,1] [,2]  
[1,] "9"  "300"  
[2,] "9"  "306"
```

Info: Working in directory C:/Users/micha/OneDrive/Desktop/BIMM 143/Class 14

Info: Writing image file hsa04110.pathview.pdf

```
keggrespathways <- rownames(keggres$greater)[1:5]  
  
# Extract the 8 character long IDs part of each string  
keggresids = substr(keggrespathways, start=1, stop=8)  
keggresids
```

```
[1] "hsa04640" "hsa04630" "hsa00140" "hsa04142" "hsa04330"
```

```
pathview(gene.data=foldchanges, pathway.id=keggresids, species="hsa")
```

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/micha/OneDrive/Desktop/BIMM 143/Class 14

Info: Writing image file hsa04640.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/micha/OneDrive/Desktop/BIMM 143/Class 14

Info: Writing image file hsa04630.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/micha/OneDrive/Desktop/BIMM 143/Class 14

Info: Writing image file hsa00140.pathview.png

'select()' returned 1:1 mapping between keys and columns

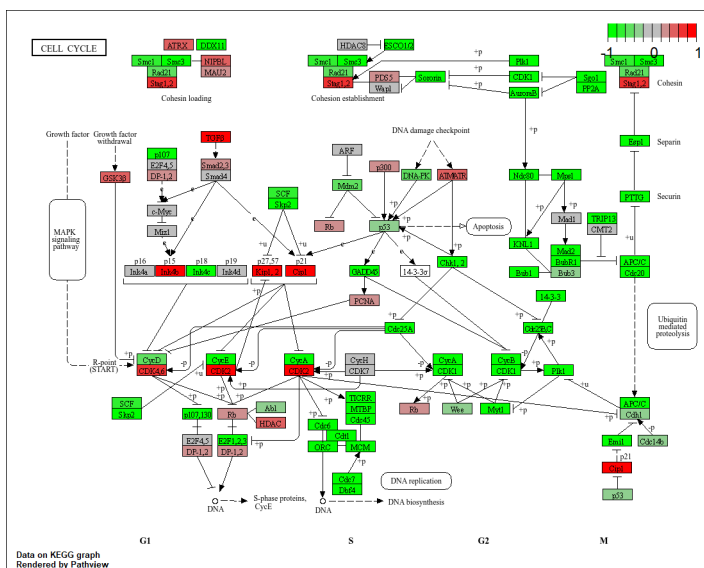
Info: Working in directory C:/Users/micha/OneDrive/Desktop/BIMM 143/Class 14

Info: Writing image file hsa04142.pathview.png

'select()' returned 1:1 mapping between keys and columns

Info: Working in directory C:/Users/micha/OneDrive/Desktop/BIMM 143/Class 14

Info: Writing image file hsa04330.pathview.png



Section 3. Gene Ontology (GO)

```

data(go.sets.hs)
data(go.subs.hs)

# Focus on Biological Process subset of GO
gobpsets = go.sets.hs[go.subs.hs$BP]

gobpres = gage(foldchanges, gsets=gobpsets)

lapply(gobpres, head)

```

\$greater

	p.geomean	stat.mean	p.val
G0:0007156 homophilic cell adhesion	8.519724e-05	3.824205	8.519724e-05
G0:0002009 morphogenesis of an epithelium	1.396681e-04	3.653886	1.396681e-04
G0:0048729 tissue morphogenesis	1.432451e-04	3.643242	1.432451e-04
G0:0007610 behavior	1.925222e-04	3.565432	1.925222e-04
G0:0060562 epithelial tube morphogenesis	5.932837e-04	3.261376	5.932837e-04
G0:0035295 tube development	5.953254e-04	3.253665	5.953254e-04
	q.val	set.size	expl
G0:0007156 homophilic cell adhesion	0.1951953	113	8.519724e-05
G0:0002009 morphogenesis of an epithelium	0.1951953	339	1.396681e-04
G0:0048729 tissue morphogenesis	0.1951953	424	1.432451e-04
G0:0007610 behavior	0.1967577	426	1.925222e-04
G0:0060562 epithelial tube morphogenesis	0.3565320	257	5.932837e-04
G0:0035295 tube development	0.3565320	391	5.953254e-04

\$less

	p.geomean	stat.mean	p.val
G0:0048285 organelle fission	1.536227e-15	-8.063910	1.536227e-15
G0:0000280 nuclear division	4.286961e-15	-7.939217	4.286961e-15
G0:0007067 mitosis	4.286961e-15	-7.939217	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.169934e-14	-7.797496	1.169934e-14
G0:0007059 chromosome segregation	2.028624e-11	-6.878340	2.028624e-11
G0:0000236 mitotic prometaphase	1.729553e-10	-6.695966	1.729553e-10
	q.val	set.size	expl
G0:0048285 organelle fission	5.841698e-12	376	1.536227e-15
G0:0000280 nuclear division	5.841698e-12	352	4.286961e-15
G0:0007067 mitosis	5.841698e-12	352	4.286961e-15
G0:0000087 M phase of mitotic cell cycle	1.195672e-11	362	1.169934e-14
G0:0007059 chromosome segregation	1.658603e-08	142	2.028624e-11
G0:0000236 mitotic prometaphase	1.178402e-07	84	1.729553e-10

\$stats

		stat.mean	exp1
G0:0007156	homophilic cell adhesion	3.824205	3.824205
G0:0002009	morphogenesis of an epithelium	3.653886	3.653886
G0:0048729	tissue morphogenesis	3.643242	3.643242
G0:0007610	behavior	3.565432	3.565432
G0:0060562	epithelial tube morphogenesis	3.261376	3.261376
G0:0035295	tube development	3.253665	3.253665

Section 4. Reactome Analysis

```
sig_genes <- res[res$padj <= 0.05 & !is.na(res$padj), "symbol"]  
print(paste("Total number of significant genes:", length(sig_genes)))
```

```
[1] "Total number of significant genes: 8147"
```

```
write.table(sig_genes, file="significant_genes.txt", row.names=FALSE, col.names=FALSE, quote=
```

Q: What pathway has the most significant “Entities p-value”? Do the most significant pathways listed match your previous KEGG results? What factors could cause differences between the two methods?

The cell cycle has the most significant “Entities p-value” of 2.63E-5. Looking at the KEGG results we see the cell cycle is also highlighted, but with a smaller p-value at 8.995727E-6. The difference come from the fact that KEGG has a smaller with more focused data set that cause any overlap to be more significant in comparison to the broader reactome dataset which causes a loss in significance.

GO Online

	Homo sapiens (REF)		upload_1 (▼ Hierarchy, NEW! ⓘ)				
GO biological process complete	#	#	expected	Fold Enrichment	±/	raw P value	FDR
trachea formation	7	7	2.78	2.52	+	1.56E-03	1.40E-02
↳anatomical structure formation involved in morphogenesis	994	543	394.94	1.37	+	2.89E-22	2.47E-20
↳anatomical structure morphogenesis	2287	1166	908.69	1.28	+	8.69E-31	1.10E-28
↳developmental process	5934	2765	2357.74	1.17	+	3.66E-37	6.41E-35
↳anatomical structure development	5359	2516	2129.28	1.18	+	9.26E-36	1.45E-33
↳animal organ morphogenesis	990	489	393.35	1.24	+	2.85E-10	9.18E-09
↳animal organ development	2930	1417	1164.17	1.22	+	1.70E-24	1.64E-22
↳trachea morphogenesis	11	10	4.37	2.29	+	6.87E-04	6.91E-03
↳trachea development	19	15	7.55	1.99	+	6.51E-04	6.60E-03
↳respiratory system development	221	140	87.81	1.59	+	1.21E-12	5.15E-11
↳system development	3655	1838	1452.23	1.27	+	4.54E-46	1.08E-43
↳multicellular organism development	4053	2047	1610.37	1.27	+	3.01E-54	1.04E-51
↳multicellular organismal process	6526	2795	2592.96	1.08	+	6.85E-10	2.10E-08
metaphase/anaphase transition of mitotic cell cycle	7	7	2.78	2.52	+	1.56E-03	1.40E-02
↳metaphase/anaphase transition of cell cycle	8	8	3.18	2.52	+	6.20E-04	6.37E-03
↳regulation of sister chromatid segregation	116	91	46.09	1.97	+	2.39E-17	1.51E-15
↳regulation of chromosome segregation	143	110	56.82	1.94	+	1.55E-19	1.12E-17
↳regulation of cell cycle process	770	489	305.94	1.60	+	8.26E-42	1.71E-39
↳regulation of cell cycle	1106	680	439.44	1.55	+	8.99E-51	2.72E-48
↳regulation of cellular process	11547	5259	4587.94	1.15	+	2.51E-83	1.92E-80
↳regulation of biological process	11979	5426	4759.59	1.14	+	2.31E-83	1.87E-80
↳biological regulation	12390	5574	4922.89	1.13	+	5.91E-81	3.90E-78